

Analytical Velocity / Yaw-Rate Limits, Envelope Geometry, and Cross-Axis Coupling of a Mecanum Platform under Steady-State LuGre Friction *and a Voltage-Limited DC-Motor Actuator (back-EMF)*

Mecanum-PINN dynamics notes

youBot / slip-spin-LuGre simulator (Julia), `:lugre adamov coupling`
 actuator model: `hybrid_ctrl/plant.jl` `MotorParams` (quasi-static DC motor)

June 5, 2026 (rev. acceleration envelope, June 6, 2026;
 rev. back-EMF actuator limits and two-gate feasibility, August 19, 2026)

Abstract

These notes collect the closed-form results derived for the four-wheel Mecanum platform of the slip-spin-LuGre simulator. We give (i) the 3-DOF equivalent (augmented) plant model the controller and observer act on; (ii) the analytical caps on the longitudinal, lateral, and yaw-rate axes and the physically distinct mechanisms that set them — *now including the DC-motor torque-speed line (back-EMF), which replaces the earlier constant-torque-cap assumption and moves both the longitudinal and the yaw caps down*; (iii) the structural shape of the no-slip / trackable velocity envelope in the translational plane ($\dot{\psi} = 0$); (iv) the extension of the caps to a conservative *velocity-acceleration* envelope built per wheel on the same friction circle (§3.5), now stated as a *two-gate* per-wheel test — friction circle *and* motor torque headroom — so that trajectory feasibility screening is coherent with the voltage-input plant the controllers actually run against; and (v) the origin and quantification of the cross-axis coupling. All quantities use the case-1 parameter set. A Word-compatible equation appendix is provided for transcription into the manuscript.

Revision note (Aug 2026). The June revision assumed a static torque cap $\tau_{\max} = 10$ N m available at any wheel speed, giving $V_{x,\text{cap}} = 4.55$ m/s and a friction-only yaw cap $\dot{\psi}_{\max} = 3.8$ rad/s. The closed-loop plant (`hybrid_ctrl/plant.jl`) is voltage-input with back-EMF: available torque falls linearly to zero at the no-load wheel speed $V_{\max}/(K_b G) = 27.73$ rad/s. The corrected caps are $V_{x,\text{cap}} \approx 1.02$ m/s and $\dot{\psi}_{\max} \approx 2.47$ rad/s (motor-limited; the friction pinch at 3.8 rad/s is kinematically unreachable). The lateral cap $V_{y,\text{crit}} \approx 0.63$ m/s is unchanged — friction still binds first there, with verified motor slack. The velocity-acceleration master envelope gains a second per-wheel inequality (the motor gate), and the wheel rotational inertia J_w , previously argued to “load the actuator, not the contact,” now enters quantitatively through that gate.

1 System and notation

The platform carries four Mecanum wheels in the O-configuration with roller-axis angles

$$\delta = \left(-\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{4}, -\frac{\pi}{4}\right), \quad \cos \delta_i = \frac{1}{\sqrt{2}} \quad \forall i, \quad \cos^2 \delta_i = \frac{1}{2}. \quad (1)$$

Wheel centres are at $p_x = (h, h, -h, -h)$, $p_y = (l, -l, l, -l)$. The body velocity is $\vec{v} = (V_x, V_y, \dot{\psi})$; the centre of mass is offset from the geometric centre by (a_X, a_Y) . The parameters used throughout are listed in Table 1 (contact / body) and Table 2 (actuator).

Sym	Value	Meaning	Sym	Value	Meaning
h	0.235 m	half-length	μ	0.5	Coulomb coeff.
l	0.15 m	half-width	$s = \mu_s / \mu_c$	1.1	stiction ratio
R	0.05 m	wheel radius	v_s	0.01 m/s	Stribeck vel.
R_a	0.0355 m	roller axle dist.	p_1	0.11	wheel viscous
$R - R_a$	0.0145 m	—	p_2	5.78×10^{-3}	roller viscous
m	30 kg	body mass	a_X	0.016 m	COM offset (long.)
m_s	35.6 kg	total mass	a_Y	-0.026 m	COM offset (lat.)
$N_{tot} = m_s g$	349.2 N	total normal	χ	{0, ..., .008}	contact-spot
J_w	5.87×10^{-3}	wheel inertia			

Table 1: Case-1 parameter set (from PlatformParams / LuGreParams). The earlier $\tau_{\max} = 10$ N m row is removed: torque is set by the motor map of Table 2, not by a static cap.

Sym	Value	Meaning	Sym	Value	Meaning
K_t	0.0335 N m/A	torque constant	$R_a^{(m)}$	2.0 Ω	winding resistance
K_b	0.0335 V s/rad	back-EMF const.	$V_{\max}$	24 V	battery bus
G	9405/364 ≈ 25.84	gear ratio	$i_{\max}$	12.8 A	current limit (near-slack)
η	0.9	gear efficiency	τ_f	0.01 N m	motor-shaft friction

Table 2: youBot base DC-motor + gearbox parameters (MotorParams, hybrid_ctrl/plant.jl, from youbot_driver/config/youbot-base.cfg). At $R_a^{(m)} = 2 \Omega$ the voltage clamp binds before the current limit ($V_{\max}/R_a^{(m)} = 12 \text{ A} < i_{\max}$), so the current limit stays slack throughout and is dropped from the analysis. Winding inductance is quasi-static ($L_a = 0$).

Contact slip kinematics. With the integrated roller-spin rate $\dot{\gamma}_i$ and the wheel rate ω_i , the per-contact slip velocities are, at the representative roller phase $\tilde{\varphi}_i \rightarrow 0$,

$$V_{px,i} = V_x - \dot{\psi} p_{y,i} - \omega_i R + \dot{\gamma}_i \sin \delta_i (R_a - R), \quad (2)$$

$$V_{py,i} = V_y + \dot{\psi} p_{x,i} + \dot{\gamma}_i \cos \delta_i (R - R_a). \quad (3)$$

The decisive structural fact is that the wheel nulls the body- x slip (its lever is $p_y = \pm l$) and the passive roller-spin state $\dot{\gamma}_i$ nulls the body- y slip (its lever is $p_x = \pm h$); the roller inertia $J_{\text{roller}} = 10^{-6}$ is driven solely by contact friction against the viscous drag p_2 .

Wheel and roller dynamics (slip retained, for reference). Away from the no-slip reduction of §2, the per-wheel and per-roller equations of Adamov & Saipulaev (their Eq. 3, lines 4–5) are

$$J_1 \ddot{\varphi}_i = M_i - p_1 \dot{\varphi}_i - F_{x,i} R, \quad (4)$$

$$J_r \ddot{\gamma}_i = -p_2 \dot{\gamma}_i - F_{x,i} \sin \delta_i (R - R_a \cos \tilde{\varphi}_i) - F_{y,i} \cos \delta_i (R_a - R \cos \tilde{\varphi}_i) + \dots, \quad (5)$$

with the platform rows (x, y, ψ) driven by $\sum_i F_{x,i}$, $\sum_i F_{y,i}$, and $\sum_i (F_{y,i} \rho_{Xi} - F_{x,i} \rho_{Yi} + M_{z,i})$. The motor enters only through the wheel equation; the roller is *passive* — $\dot{\gamma}_i$ is sustained only by contact terms against p_2 . This actuated-wheel / passive-roller asymmetry splits the caps in §3.

DC-motor actuator model (what M_i actually is). The closed-loop plant is *voltage*-input. With ZOH bus voltage V_i (slew-limited, $|V_i| \leq V_{\max}$) the quasi-static motor map evaluated every RHS call of the plant is

$$i = \frac{V_i - K_b G \omega_i}{R_a^{(m)}}, \quad M_i = G \eta (K_t i - \tau_f \text{sign } \omega_i), \quad (6)$$

so the torque available to wheel i at speed ω_i with the voltage at its cap is the *torque-speed line*

$$\tau_{avail}(\omega) = \underbrace{G \eta \left(\frac{K_t V_{\max}}{R_a^{(m)}} - \tau_f \right)}_{\tau_{stall} \approx 9.12 \text{ N m}} - \underbrace{\frac{G^2 \eta K_t K_b}{R_a^{(m)}}}_{\approx 0.337 \text{ N m s/rad}} |\omega|. \quad (7)$$

Two landmarks organise everything that follows. The *stall torque* $\tau_{stall} \approx 9.12 \text{ N m}$ (not 10 N m : the $24 \text{ V} / 2 \Omega$ voltage clamp binds before the 12.8 A current limit) is the most the motor can ever deliver, at standstill. The *no-load speed*

$$\omega_{nl} = \frac{V_{\max}}{K_b G} \approx 27.73 \text{ rad/s} \quad (\approx 27.04 \text{ with } \tau_f) \quad (8)$$

is the wheel speed at which back-EMF consumes the whole bus and *no torque exists at all* — a hard kinematic wall no control law can cross, independent of friction. In body terms ω_{nl} corresponds to $V_x = 1.386 \text{ m/s}$, $V_y = 1.386 \text{ m/s}$, or $\dot{\psi} = \omega_{nl} R / (l + h) = 3.60 \text{ rad/s}$ of single-axis motion. This is the same wall the v3 trajectory-tier screening applies (`trajsets.jl`: max per-wheel speed against 27.73 rad/s , with $\gtrsim 85\%$ of no-load measured catastrophic for every controller).

Steady-state friction. The LuGre/Stribeck steady force per contact is a bounded vector aligned with the slip,

$$\vec{F}_i = -N_i g_t(\|V_p\|) \frac{V_{p,i}}{\|V_{p,i}\|}, \quad g_t = \mu_c \left[1 + (s - 1) e^{-(\|V_p\|/v_s)^2} \right], \quad (9)$$

so $\|\vec{F}_i\|$ ranges from $N_i \mu_s$ (static) down to $N_i \mu_c$ (kinetic): each contact has an essentially isotropic friction circle of radius $\sim N_i \mu$.

Roller-frame force decomposition. Every contact force lives in the plane of its roller, so the natural basis is the roller frame — the axle direction $\hat{e}_i = (\cos \delta_i, \sin \delta_i)$ and the rolling direction $\hat{n}_i = (-\sin \delta_i, \cos \delta_i)$ — in which

$$\vec{F}_i = F_{\parallel,i} \hat{e}_i + F_{\perp,i} \hat{n}_i, \quad F_{\parallel,i}^2 + F_{\perp,i}^2 \leq (\mu N_i)^2. \quad (10)$$

The two components have distinct owners. Along the axle the roller cannot roll, so $F_{\parallel}$ is the wheel traction set by the motor (carried with the bearing p_1). Along the rolling direction the roller spins freely against p_2 , so $F_{\perp} = p_2 \dot{\gamma}_i / (R - R_a)$ is the roller-sustaining drag, supplied by contact friction because the roller is passive. This is the p_1/p_2 split made geometric: $F_{\parallel} \leftrightarrow \text{motor}$, $F_{\perp} \leftrightarrow \text{contact}$. A longitudinal command leaves $F_{\perp} = 0$ and the circle empty (a wheel/motor limit); a lateral or yaw command forces $F_{\perp} \neq 0$ and fills it (a roller/contact limit).

The motor sees the body- x component. The wheel equation reacts the body- x contact force, which in the roller frame is

$$F_{x,i} = F_{\parallel,i} \cos \delta_i - F_{\perp,i} \sin \delta_i = \frac{1}{\sqrt{2}}(F_{\parallel,i} \mp F_{\perp,i}), \quad (11)$$

so the per-wheel actuator balance the plant enforces is

$$M_i = \underbrace{R(F_{\parallel,i} \cos \delta_i - F_{\perp,i} \sin \delta_i)}_{\text{contact reaction}} + p_1 \omega_i + J_w \dot{\omega}_i \leq \tau_{avail}(\omega_i). \quad (12)$$

Note that $F_{\perp}$ — nominally the “friction-side” component — has a body- x component and therefore *spends motor torque*: the friction circle and the actuator line are two coupled gates, not two independent ones. This coupling is what moves the yaw cap (§3.3).

2 Equivalent (augmented) plant model

The controller and disturbance observer are built on a 3-DOF rigid-body equivalent in which the wheel/roller rotational inertias are reflected into the body and the lumped contact dissipation is written as equivalent viscous and Coulomb terms. With $\vec{v} = (V_x, V_y, \dot{\psi})$,

$$M_{aug} \ddot{\vec{v}} + C(\dot{\psi}) \dot{\vec{v}} + D_{eq} \vec{v} + \vec{f}_c(\vec{v}) = \vec{F}_p. \quad (13)$$

The augmented mass matrix reflects J_w through the wheel kinematics and keeps the COM coupling,

$$M_{aug} = \begin{bmatrix} \tilde{m} & 0 & -ma_Y \\ 0 & \tilde{m} & ma_X \\ -ma_Y & ma_X & I_\psi \end{bmatrix}, \quad \tilde{m} = m_s + \frac{4J_w}{R^2}, \quad I_\psi = I_s + \frac{4(l+h)^2}{R^2} J_w. \quad (14)$$

The gyroscopic/centrifugal vector (the only translation–yaw coupling once M_{aug} is formed) is

$$C(\dot{\psi}) \vec{v} = \begin{bmatrix} -m_s \dot{\psi} V_y - ma_X \dot{\psi}^2 \\ m_s \dot{\psi} V_x - ma_Y \dot{\psi}^2 \\ m \dot{\psi} (a_X V_x + a_Y V_y) \end{bmatrix}. \quad (15)$$

Equivalent damping from the no-slip (Model N) reduction. In Adamov & Saipulaev the dissipation lives only in the per-wheel/roller equations — base–wheel joint $p_1 \dot{\phi}_i$ and wheel–roller joint $p_2 \dot{\gamma}_i$ — and the platform equations carry no explicit body damping. Imposing the no-slip constraints $V_{PiX} = V_{PiY} = 0$ at $\tilde{\phi}_i = 0$ and solving,

$$\dot{\gamma}_i = -\frac{V_y + \rho_{Xi} \dot{\psi}}{\cos \delta_i (R - R_a)}, \quad \dot{\phi}_i = \frac{1}{R} [V_x - \rho_{Yi} \dot{\psi} + (V_y + \rho_{Xi} \dot{\psi}) \tan \delta_i]. \quad (16)$$

With the Rayleigh dissipation $\mathcal{F} = \frac{1}{2} \sum_i (p_1 \dot{\phi}_i^2 + p_2 \dot{\gamma}_i^2)$ the equivalent body damping is the Hessian $D_{eq} = \sum_i (p_1 \nabla \dot{\phi}_i \nabla \dot{\phi}_i^\top + p_2 \nabla \dot{\gamma}_i \nabla \dot{\gamma}_i^\top)$, which (all off-diagonals cancelling) is

$$D_{eq} = \text{diag} \left(\underbrace{\frac{4p_1}{R^2}}_{d_x}, \underbrace{\frac{4p_1}{R^2} + \frac{8p_2}{(R-R_a)^2}}_{d_y}, \underbrace{\frac{4p_1(l+h)^2}{R^2} + \frac{8p_2 h^2}{(R-R_a)^2}}_{d_\psi} \right) \approx \text{diag}(176, 396, 38). \quad (17)$$

The p_1/p_2 split that drives every cap. Each entry separates into a wheel-joint (p_1) part paid by the motors (at steady state $M_i = F_{X_i}R + p_1\dot{\phi}_i$, so bearing drag is covered by motor torque and never loads the circle) and a roller-joint (p_2) part that must be supplied by contact friction because the roller is passive. Consequently every friction-limited cap (§3) is set by the p_2 part of its axis, while the p_1 part sets the actuator/power demand: V_x has no p_2 term (motor-limited), whereas V_y and $\dot{\psi}$ are governed by their p_2 parts (friction-limited) — *but only up to the motor gate* (12): with the torque–speed line (7) in place, V_x is motor-limited outright and $\dot{\psi}$ turns out to be motor-limited too, the friction pinch lying above what the actuator can reach (§3.3). The control wrench is the wheel-torque map

$$\vec{F}_p = \left(\frac{1}{R} \sum_i \tau_i, \frac{1}{R}(-\tau_1 + \tau_2 + \tau_3 - \tau_4), \frac{l+h}{R}(-\tau_1 + \tau_2 - \tau_3 + \tau_4) \right). \quad (18)$$

3 Velocity and yaw-rate caps

3.1 Longitudinal cap — motor torque–speed line vs bearing drag ($\dot{\psi} = 0$)

At constant V_x the rollers need not spin ($\dot{\gamma}_i = 0$, hence $F_{\perp,i} = 0$), and a wheel rolling at steady speed needs no traction ($F_{\parallel,i} \rightarrow 0$): the circle is empty. The steady resistance the motor must hold is the bearing torque $p_1\omega_i$ alone, and it must hold it *from the torque–speed line* (7), not from a speed-independent cap. The cap is the intersection $\tau_{avail}(\omega) = p_1\omega$:

$$\omega_* = \frac{\tau_{stall}}{p_1 + G^2\eta K_t K_b / R_a^{(m)}} = \frac{9.12}{0.11 + 0.337} \approx 20.39 \text{ rad/s}, \quad \boxed{V_{x,cap} = \omega_* R \approx 1.02 \text{ m/s}} \quad (19)$$

(current draw at the cap ≈ 3.2 A, far under $i_{\max}$ — the voltage clamp and back-EMF, never the current limit, set this axis). This is a *soft* cap: the circle never binds. Above it lies the hard no-load wall (8) at $V_x = 1.386$ m/s where torque authority vanishes identically. As a fraction of no-load, $V_{x,cap}$ sits at $\omega_*/\omega_{nl} \approx 73.5\%$ — squarely inside the band the v3 feasibility screening measured (everything $\leq 76\%$ of no-load flyable, $\geq 85\%$ catastrophic for every controller; target $\leq 70\%$ for margin). The corrected analytical cap and the empirical screen are the same wall.

Superseded value. The June revision gave $V_{x,cap} = \tau_{\max}R/p_1 = 10 \times 0.05/0.11 \approx 4.55$ m/s, assuming a static 10 N m torque cap at any speed. That figure is unreachable: it exceeds the no-load ceiling 1.386 m/s by $3.3\times$. The stall torque $\tau_{stall} \approx 9.12$ N m shows the old number was only ever approximately right *at standstill*, where it overestimates even that by $\sim 10\%$.

Traction headroom vs speed. What remains for acceleration at speed V_x is

$$F_{head}(V_x) = \frac{\tau_{avail}(V_x/R) - p_1 V_x/R}{R} = 182.3 - 66.0 V_x \text{ [N/wheel]}, \quad (20)$$

falling from 182 N/wheel at rest (friction binds, $\mu N_3 = 34.8$ N) through μN_3 at $V_x \approx 0.83$ m/s (59.5% of no-load) to zero at the cap. Below ≈ 0.8 m/s the friction circle is the longitudinal constraint; above it the motor is. The transient (maneuver) limit at low speed remains the aggregate friction circle,

$$a_{x,\max} = \frac{(\sum_i N_i)\mu}{m_s} = \frac{349.2 \times 0.5}{35.6} \approx 4.9 \text{ m/s}^2, \quad (21)$$

per-wheel-pinned to 2.93 m/s^2 at rest (§3.5) and shrinking with speed per (20).

3.2 Lateral cap — friction-circle limited ($\dot{\psi} = 0$), motor slack verified

For pure strafe the rollers must spin to null $V_{py,i}$, at $\dot{\gamma}_i = -V_y/(\cos \delta_i(R - R_a))$, so each contact carries the roller-sustaining drag

$$F_{\perp,i} = \frac{p_2 \dot{\gamma}_i}{R - R_a} = \frac{\sqrt{2} p_2}{(R - R_a)^2} V_y \quad (\text{along } \hat{n}_i). \quad (22)$$

This $F_{\perp}$ is the same on every wheel and sums to a pure $-y$ drag $2\sqrt{2} F_{\perp}$. The steady balance forces the tractions to supply this in y and nothing net in x , which — by the symmetric $\pm$ pattern — pins $|F_{\parallel,i}| = F_{\perp,i}$ at every wheel. The two roller-frame components being equal, each wheel's resultant has magnitude

$$\|\vec{F}_i\| = \sqrt{2} F_{\perp,i} = \frac{2p_2}{(R - R_a)^2} V_y \approx 55.0 V_y \text{ [N]}, \quad (23)$$

directed along $\pm \text{body-}x$. Saturation occurs when $\sqrt{2} F_{\perp,i} = N_i \mu$; with $N_{\min} = N_3 \approx 69.6 \text{ N}$ the lightest wheel saturates first:

$$V_{y,crit} = \frac{N_{\min} \mu (R - R_a)^2}{2p_2} \approx 0.63 \text{ m/s}, \quad (24)$$

$$V_{y,agg} = \frac{(\sum_i N_i) \mu (R - R_a)^2}{8p_2} \approx 0.79 \text{ m/s}. \quad (25)$$

Since $V_{y,crit} \propto \mu$, $V_{y,crit}(\mu) \approx 1.265 \mu = \{0.51, 0.63, 0.76\} \text{ m/s}$ for $\mu = \{0.4, 0.5, 0.6\}$ — the cleanest plant-vs-controller discriminator — and the Stribeck dip places the onset in the hysteretic band $[V_{y,crit}, sV_{y,crit}] \approx [0.63, 0.70] \text{ m/s}$.

Motor check. At $V_y = 0.63 \text{ m/s}$ the wheels turn at $|\omega_i| = V_y/R = 12.6 \text{ rad/s}$ (45% of no-load) and each motor holds $F_{x,i} = \pm \sqrt{2} F_{\perp}$ through (12): $M_i = \sqrt{2} F_{\perp} R + p_1 \omega_i \approx 3.13 \text{ N m}$ against $\tau_{avail}(12.6) \approx 4.85 \text{ N m}$ — $\sim 1.7 \text{ N m}$ of slack. The no-load wall for strafe is $V_y = 1.386 \text{ m/s}$, more than twice the friction cap. The lateral axis therefore remains friction-limited exactly as before; the motor gate is recorded here because combined strafe-spin states (§3.5) load both gates at once.

3.3 Yaw-rate cap: the motor gate binds before the friction circle

A pure spin moves each contact tangentially. Its body- y component $\dot{\psi} \rho_{Xi} = \pm \dot{\psi} h$ is rolled out by the passive roller at $|\dot{\gamma}_i| = \sqrt{2} h \dot{\psi} / (R - R_a)$, costing

$$F_{\perp,i} = \frac{p_2 \dot{\gamma}_i}{R - R_a} = \frac{\sqrt{2} p_2 h}{(R - R_a)^2} \dot{\psi} \approx 9.1 \dot{\psi} \text{ N}, \quad (26)$$

the same passive force that set $V_{y,crit}$, now driven by the long spin lever h . It loads the circle directly. The roller-joint dissipation $d_{\dot{\psi}}^{(p_2)} = 8p_2 h^2 / (R - R_a)^2 \approx 12.2$ is the power coefficient; the *contact* yaw moment of $F_{\perp}$ is only

$$\sum_i F_{\perp,i} \ell_{\perp,i} = \frac{4p_2 h (h - l)}{(R - R_a)^2} \dot{\psi} \approx 2.2 \dot{\psi} \text{ N m}, \quad (27)$$

at the short lever $\ell_\perp = (h - l)/\sqrt{2}$; the rest of $d_\psi^{(p_2)}$ is fed through the wheels and never sits on the circle. With the traction headroom

$$|F_{\parallel,i}| \leq \sqrt{(\mu N_i)^2 - F_{\perp,i}^2}, \quad (28)$$

the friction band collapses as $F_\perp \rightarrow \mu N_i$, the lightest wheel pinching first at

$$\dot{\psi}_{fric} = \frac{\mu N_{\min}(R - R_a)^2}{\sqrt{2} p_2 h} = \frac{\sqrt{2} V_{y,crit}}{h} \approx 3.8 \text{ rad/s}. \quad (29)$$

But this pinch is never reached. A pure spin at rate $\dot{\psi}$ demands wheel speeds $|\omega_i| = (l + h)\dot{\psi}/R = 7.7\dot{\psi}$ from (16), so the no-load wall (8) alone already caps the spin at $\dot{\psi} < \omega_{nl}R/(l + h) = 3.60$ rad/s, below (29). The torque balance (12) binds earlier still. At steady spin the min-norm traction holding the contact yaw moment (27) is $|F_{\parallel,i}| \approx 2.02\dot{\psi}$, which *partially cancels* $F_\perp$ in the body- x reaction (11): $|F_{x,i}| = (9.1 - 2.02)\dot{\psi}/\sqrt{2} \approx 5.0\dot{\psi}$ N. The motor gate reads

$$\underbrace{(0.05 \times 5.0 + 0.11 \times 7.7)}_{1.10 \text{ N ms}} \dot{\psi} \leq \tau_{avail}(7.7\dot{\psi}) = 9.12 - 2.60\dot{\psi}, \quad (30)$$

giving the corrected, motor-limited yaw cap

$$\boxed{\dot{\psi}_{\max} = \frac{\tau_{stall}}{1.10 + 2.60} \approx 2.47 \text{ rad/s}} \quad (68.5\% \text{ of no-load}). \quad (31)$$

At this rate the friction side is slack: $F_\perp = 22.5$ N against a headroom $\sqrt{(\mu N_3)^2 - F_\perp^2} \approx 26.5$ N for the $|F_\parallel| \approx 5.0$ N actually used. The hierarchy on this axis is therefore $\dot{\psi}_{\max}(2.47) < \dot{\psi}_{nl}(3.60) < \dot{\psi}_{fric}(3.8)$: motor first, then the kinematic wall, then a friction pinch the hardware cannot reach. This matches the observed failure of high-spin trajectories (e.g. `spiral_orbit` combo 37 at $\dot{\psi} \approx 2.04$ with strafe, 85% of no-load, catastrophic for all four controllers) far better than a 3.8 rad/s friction story does.

3.4 Acceleration headroom and the velocity–acceleration envelope: the two-gate test

The caps of §3.1–3.3 are the $\dot{\vec{v}} = 0$ slice of a larger reachable set. To bound transient maneuvers we ask, at a velocity $\vec{v}$ already inside the envelope, how much acceleration $\dot{\vec{v}}$ the contacts can still supply on the *same* friction circle *and* how much the motors can still deliver on the *same* torque–speed line. The roller-frame split (10) answers the first half: $F_{\perp,i}(\vec{v})$ is velocity-slaved and consumes the circle radially, while $F_{\parallel,i}$ — the motor/traction component — must carry the entire inertial demand and lives in whatever headroom remains. Eq. (28) is therefore not just the yaw-cap pinch but a general per-wheel statement — and the motor gate (12) is its actuator-side twin.

Per-wheel resolution. Write the bare-body balance (the contact form of (14), forces explicit) with the velocity-known drag moved to the right,

$$A_\parallel \vec{F}_\parallel = \underbrace{M \dot{\vec{v}} + C(\dot{\psi}) \vec{v} - A_n \vec{F}_\perp(\vec{v})}_{\vec{b}(\vec{v}, \dot{\vec{v}})}, \quad (32)$$

where the allocators carry each wheel's roller-frame force into the body wrench (F_x, F_y, M_z) ,

$$A_{\parallel} = \begin{bmatrix} \cos \delta_i \\ \sin \delta_i \\ \rho_{X_i} \sin \delta_i - \rho_{Y_i} \cos \delta_i \end{bmatrix}_i^{\top}, \quad A_n = \begin{bmatrix} -\sin \delta_i \\ \cos \delta_i \\ \rho_{X_i} \cos \delta_i + \rho_{Y_i} \sin \delta_i \end{bmatrix}_i^{\top}, \quad (33)$$

whose moment rows are the roller-frame arms $\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}\lambda_i$, $\ell_{\perp,i} = \frac{h-l}{\sqrt{2}}\nu_i$ of (51). Numerically,

$$A_{\parallel} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -0.385 & 0.385 & -0.385 & 0.385 \end{bmatrix}, \quad A_{\parallel}^+ = \begin{bmatrix} 0.354 & -0.354 & -0.918 \\ 0.354 & 0.354 & 0.918 \\ 0.354 & 0.354 & -0.918 \\ 0.354 & -0.354 & 0.918 \end{bmatrix}. \quad (34)$$

Min-norm allocation. $A_{\parallel}$ is 3×4 with $\ker A_{\parallel} = \frac{1}{2}(1, 1, -1, -1)$ — the front/rear “squeeze” that produces zero body wrench but loads all four circles equally *and* costs motor torque on every wheel, so it is doubly undesirable. The feedforward injects none of it, so we take the min-norm allocation $\vec{F}_{\parallel} = A_{\parallel}^+ \vec{b}$, which gives the largest envelope; a factor of safety enters separately as $\mu N_i \rightarrow s \mu N_i$, $s \in [0.7, 0.8]$, shrinking the $F_{\perp}$ caps and $F_{\parallel}$ headroom together.

Bare vs. augmented mass — and where the wheel inertia went. The contact-force balance (32) uses the *bare* body mass $M = \text{diag-block}(m_s, m_s, I_s)$ with the COM coupling, *not* the augmented M_{aug} of (14). The $4J_w/R^2$ and $4(l+h)^2 J_w/R^2$ reflections are wheel rotational inertia, spun up by motor torque through the wheel equation $J_w \ddot{\phi}_i = M_i - p_1 \dot{\phi}_i - F_{x,i} R$; they load the actuator, not the contact. Using M_{aug} in (32) would double-count the wheel inertia onto the circle and spuriously shrink the acceleration envelope by $\tilde{m}/m_s \approx 1.26$ on translation and $I_{\psi}/I_s \approx 1.31$ on yaw. *The wheel inertia does not vanish, however — it re-enters through the motor gate of (35) below as $J_w \dot{\omega}_i$, which is precisely the “loads the actuator” clause made quantitative.* A feasibility test that checks only the circle silently drops it; the two-gate test keeps each inertia on its own side: $F_{\parallel}$ sees m_s ; the motor sees $\tilde{m}$.

The envelope (two per-wheel gates). The reachable set is the intersection of the per-wheel friction headroom law (the acceleration generalization of (28)) *and* the per-wheel actuator balance, both evaluated on the same allocation $\vec{F}_{\parallel} = A_{\parallel}^+ \vec{b}$:

$$\boxed{\begin{array}{ll} \text{(friction)} & (A_{\parallel}^+ \vec{b})_i^2 + F_{\perp,i}(\vec{v})^2 \leq (\mu N_i)^2, \\ \text{(motor)} & R \left[(A_{\parallel}^+ \vec{b})_i \cos \delta_i - F_{\perp,i}(\vec{v}) \sin \delta_i \right] + p_1 \omega_i + J_w \dot{\omega}_i \leq \tau_{avail}(\omega_i), \end{array} \quad i = 1 \dots 4,} \quad (35)$$

with the no-slip wheel kinematics from (16) (and its time derivative, same sign pattern),

$$\omega_i = \frac{V_x + \tan \delta_i V_y + (\rho_{X_i} \tan \delta_i - \rho_{Y_i}) \dot{\psi}}{R} = \frac{V_x \pm V_y \pm (l+h) \dot{\psi}}{R}, \quad (36)$$

$$\dot{\omega}_i = \frac{a_x \pm a_y \pm (l+h) \alpha}{R}, \quad \dot{\vec{v}} = (a_x, a_y, \alpha), \quad (37)$$

and τ_{avail} from (7). The velocity-slaved drag fixes $F_{\perp}$ (radial, friction side; body- x projection, motor side); the inertial demand sets $F_{\parallel}$ (residual on both gates); the envelope shrinks to a point

as $\vec{v}$ approaches the velocity caps — the friction edge on the lateral axis, the motor edge on the longitudinal and yaw axes. The drag-relocation vector $\Phi(\vec{v}) = A_n \vec{F}_\perp$ is velocity-only and linear,

$$\Phi(\vec{v}) = \left(0, \frac{4p_2}{(R-R_a)^2} V_y, \frac{4p_2 h(h-l)}{(R-R_a)^2} \dot{\psi} \right) = (0, 110.0 V_y, 2.20 \dot{\psi}), \quad (38)$$

its yaw entry exactly the contact yaw moment $2.2 \dot{\psi}$ of (27) and its lateral entry the net strafe drag $2\sqrt{2}F_\perp$ of §3.2 — the velocity envelope re-entering as the affine offset of the acceleration problem. (The June revision printed the symbolic coefficients of (38) as $2\sqrt{2}p_2/(R-R_a)^2$ and $2\sqrt{2}p_2 h(h-l)/(R-R_a)^2$; those forms are typographical — they evaluate to 77.8 and 1.55 — while the numbers 110.0 and 2.20 it quoted, reproduced here, are correct.)

Acceleration intercepts. At rest ($\vec{v} = 0$, $F_\perp = 0$, the full circle free, stall torque available) the pure-axis caps from (35) are

$$a_{x,cap} = 2.93, \quad a_{y,cap} = 2.86 \text{ m/s}^2, \quad \ddot{\psi}_{cap} = 9.62 \text{ rad/s}^2, \quad (39)$$

all pinched at the rear-left wheel ($N_{\min}$) on the *friction* gate — the motor gate is slack at rest, 182 N/wheel of stall traction against the $\mu N_3 = 34.8$ N that binds — and all below the aggregate isotropic value $\mu g = 4.9 \text{ m/s}^2$, because a single wheel binds first under any feasible allocation. These intercepts are unchanged from the friction-only analysis; what changes is the *shape at speed*: the friction gate shrinks the envelope radially as $F_\perp$ grows, and the motor gate shrinks it further wherever wheel speeds are high (per (20), longitudinal traction headroom falls from 92.9 N/wheel at $V_x = 0.5$ to 3.5 N/wheel at $V_x = 1.0$ m/s).

Binding-wheel reduction (gating form). Across the operating box the rear-left wheel (#3, $\mu N_3 = 34.8$ N) is the binding *friction* constraint in every pure direction and in most mixed states, while the binding *motor* constraint is motion-dependent — the wheel with the largest $|\omega_i|$ plus the largest body- x reaction, not necessarily #3. The four-wheel test (35) therefore collapses to a scalar gate pair, evaluated on the candidate binding wheel of each side. With $\rho_{X,3} = -h$, $\rho_{Y,3} = +l$, and the third row of $A_\parallel^+ = (0.354, 0.354, -0.918)$,

$$(\text{friction, wheel 3}) \quad F_{\perp,3} = \kappa (V_y - h\dot{\psi}), \quad \kappa = \frac{\sqrt{2}p_2}{(R-R_a)^2} = 38.9, \quad (40)$$

$$F_{\parallel,3} = 0.354 (b_x + b_y) - 0.918 b_\Omega, \quad (41)$$

$$\text{feasible} \iff F_{\parallel,3}^2 \leq (\mu N_3)^2 - F_{\perp,3}^2, \quad (42)$$

$$(\text{motor, wheel } j = \arg \max_i |\omega_i|) \quad M_j = R \frac{1}{\sqrt{2}} (F_{\parallel,j} \mp F_{\perp,j}) + p_1 \omega_j + J_w \dot{\omega}_j, \quad (43)$$

$$\text{feasible} \iff M_j \leq \tau_{avail}(\omega_j) \quad \text{and} \quad |\omega_i| < \omega_{nl} \quad \forall i, \quad (44)$$

with $\vec{b}$ from (32) expanded as

$$b_x = m_s a_x - m a_Y \alpha - m_s \dot{\psi} V_y - m a_X \dot{\psi}^2, \quad (45)$$

$$b_y = m_s a_y + m a_X \alpha + m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 - 110.0 V_y, \quad (46)$$

$$b_\Omega = -m a_Y a_x + m a_X a_y + I_s \alpha + m \dot{\psi} (a_X V_x + a_Y V_y) - 2.20 \dot{\psi}, \quad (47)$$

($\dot{\vec{v}} = (a_x, a_y, \alpha)$). Only in the off-axis corners (large $|V_y|$ against opposing spin) does the front-left wheel (#1) bind first on the friction side; the sampler should carry the #1 row as a second friction

gate there, or fall back to the full (35). On the motor side the no-load wall $|\omega_i| < 27.73$ rad/s is itself the cheapest first screen — it is the criterion the v3 trajectory tiers apply, and it rejects every state the torque balance would reject at zero load. A coarse 729-node lookup of the maximum single-axis acceleration over the velocity grid accompanies these notes for rejection sampling; with the motor gate added, the lookup must be regenerated to evaluate *both* gates.

3.5 Out-of-plane load transfer: the bound that sets the operating acceleration cap

The friction-circle caps of §3.4 ($a_{cap} \approx 2.9$ m/s²) are derived in the planar 3-DOF model, where the per-wheel normal loads N_i are *static* (set once by weight and the COM offset, Table 1). A platform accelerating at a redistributes those loads through the out-of-plane pitch/roll the planar model omits. For a rigid (suspension-free) body the quasi-static per-wheel transfer is

$$\Delta N_i = \frac{m a z_{com}}{4d}, \quad d = \begin{cases} h & \text{longitudinal} \\ l & \text{lateral,} \end{cases} \quad (48)$$

with z_{com} the COM height and $2h, 2l$ the wheelbase/track. Each contact force is *linear* in N_i ($\|\vec{F}_i\| \leq \mu N_i$), so $\Delta N_i/N_i$ is exactly the fractional per-wheel force error the planar dataset cannot reproduce. The lateral axis binds ($l < h$) and the lightest wheel #3 ($N_3 = 69.6$ N, the friction-circle binding wheel of §3.3) is the worst case.

Holding this neglected effect below 5% bounds the operating acceleration. The plant carries no z -DOF, so z_{com} is a free parameter; for the youBot we take $z_{com} = R = 0.05$ m (COM at hub height), with $z_{com} = 2R$ as a sensitivity:

$$a_{5\%}^{lat} = \frac{0.05 \cdot 4l N_3}{m z_{com}} = \begin{cases} 1.39 \text{ m/s}^2 & z_{com} = R, \\ 0.70 \text{ m/s}^2 & z_{com} = 2R. \end{cases} \quad (49)$$

a [m/s ²]	$z_{com} = R$		$z_{com} = 2R$	
	$\Delta N/\bar{N}$	$\Delta N/N_3$	$\Delta N/\bar{N}$	$\Delta N/N_3$
0.5	1.4%	1.8%	2.9%	3.6%
1.0	2.9%	3.6%	5.7%	7.2%
1.5	4.3%	5.4%	8.6%	10.8%
2.0	5.7%	7.2%	11.5%	14.4%

Table 3: Per-wheel lateral load-transfer fraction (48) ($\bar{N} = 87.3$ N mean, $N_3 = 69.6$ N binding wheel).

We therefore cap the operating envelope at $a \leq 1.0$ m/s², well inside the 2.9 m/s² friction-circle limit. At $z_{com} = R$ this holds the worst-wheel transfer to 3.6% (longitudinal 1.8%); at $z_{com} = 2R$ it reaches 7.2%, so $a \leq 1.0$ is the appropriate, COM-height-aware cap for the youBot’s low ($z_{com} \lesssim R$) centre of mass. This is the operational guarantee that the planar Julia dataset — and the high-fidelity engine benchmark validating it — never enters a regime where neglected load transfer exceeds the 5% fidelity budget. Note the motor gate does not relax this bound: it is *additional* to it, and at high wheel speed it is the tighter of the two (§3.4, traction headroom $\rightarrow 0$ as $V_x \rightarrow 1.02$ m/s).

4 Bounding envelope geometry ($\dot{\psi} = 0$)

The trackable-velocity set in the translational plane is neither a rhombus nor an ellipse. Each contact enforces an isotropic circle $\|\vec{F}_i\| \leq N_i\mu$; since the slip components are affine in (V_x, V_y) , each wheel's limit is an ellipse and the feasible set is the intersection of four oriented ellipses — a rounded convex region, *further clipped on the V_x side by the actuator*. The two axes are bounded by different mechanisms — a near-vertical wall in V_x at the motor drag-line intersection (19) (soft edge ± 1.02 m/s inside the hard no-load wall ± 1.39 m/s), a rounded friction fold in V_y at ± 0.63 m/s — breaking the 90° symmetry an ellipse would require. A compact description is an anisotropic superellipse

$$\left| \frac{V_x}{V_{x,cap}} \right|^n + \left| \frac{V_y}{V_{y,crit}} \right|^n = 1, \quad 1 < n < 2, \quad \frac{V_{x,cap}}{V_{y,crit}} \approx 1.6, \quad (50)$$

with the diamond-like skeleton ($n \rightarrow 1$) from the $\pm 45^\circ$ roller directions and the rounding ($n > 1$) from the LuGre smoothing. The anisotropy is far milder than the friction-only analysis suggested ($V_{x,cap}/V_{y,crit} \approx 7$ under the static-torque assumption): the back-EMF wall pulls the longitudinal edge in by $4.5\times$, and the platform is closer to isotropic in *reachable* speed than in *limiting mechanism*. The envelope is three-dimensional: the coupled LuGre–Adamov law folds the spin $\omega_{z,i} = \dot{\psi}$ into the slip measure, so rising $\dot{\psi}$ eats the translational budget and the body is pinched in the combined-motion corners — the same pinch that §3.4 quantifies on the acceleration side, now with the motor gate (35) as the second, usually tighter, spin-side constraint ($\dot{\psi}_{\max} = 2.47$ rad/s against the friction pinch's unreachable 3.8).

5 Cross-axis coupling

5.1 Coupling channel

The off-centre COM makes M_{aug} non-diagonal, so a force residual on one axis leaks into the others through the off-diagonal entries of M_{aug}^{-1} (nonzero iff $a_X, a_Y \neq 0$). The per-wheel yaw contribution is $M_{z,i} = p_{x,i}F_{y,i} - p_{y,i}F_{x,i} + M_{z,i}^{spin}$.

5.2 Lateral axis (dominant)

In the roller frame $M_{z,i} = F_{\parallel,i}\ell_{\parallel,i} + F_{\perp,i}\ell_{\perp,i}$ with

$$\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}\lambda_i \ (\lambda = (-, +, -, +)), \quad \ell_{\perp,i} = \frac{h-l}{\sqrt{2}}\nu_i \ (\nu = (+, +, -, -)), \quad (51)$$

so the traction moment acts at the long arm 0.272 m and the roller-drag moment at the short arm 0.060 m — a $4.5\times$ leverage advantage for $F_{\parallel}$. The moment is exactly zero up to and including $V_{y,crit}$ (load-independent symmetric forces) and ramps from zero only above it,

$$M_z^{lat} = \mu mg a_X \frac{l}{h} \approx 1.50 \text{ N m} \longrightarrow \mu mg a_X \frac{h+l}{\sqrt{2}h} \approx 2.7 \text{ N m}, \quad (52)$$

as the roller-spin cap clips the weak-lever cancelling term. This routes the front–rear a_X imbalance through $F_{\parallel}$, which — via M_{aug}^{-1} — contaminates V_x .

5.3 Longitudinal axis (present but subdominant)

Longitudinal drive needs no roller spin ($F_{\perp,i} = 0$), so each saturated force lies along its axle at the full traction arm with no cancelling term:

$$M_z^{lon} = \mu m g a_Y \frac{h+l}{\sqrt{2}l} \approx 6.94 \text{ N m.} \quad (53)$$

Its bare magnitude is the largest of the three, yet the axis is benign operationally: V_x friction-saturates only transiently, the circle is otherwise empty, and the kick is self-limiting. (Under the motor model the longitudinal axis now hits its *actuator* wall at 1.02 m/s long before any friction saturation — the soft cap of §3.1 — so this transient moment is if anything rarer than the friction-only picture implied.)

5.4 Yaw-rate axis: centrifugal creep coupling (benign)

The centrifugal terms in (15) are a body force $\vec{F}_{cf} = m\dot{\psi}^2(a_X, a_Y)$, $\|\vec{F}_{cf}\| = m|a|\dot{\psi}^2$, $|a| = 0.031$ m, held by the translational friction until $\dot{\psi}_{cf} = \sqrt{\mu N_{tot}/(m|a|)} \approx 14$ rad/s. Because $\dot{\psi}_{cf} \gg \dot{\psi}_{\max}$ the platform cannot reach centrifugal gross slip. Below the cap the coupling is a small steady creep $v_{creep} \approx m|a|\dot{\psi}^2/d_t \approx 1.2$ cm/s at $\dot{\psi} \approx 2$. Yaw remains the most benign axis.

5.5 Role of the COM offset: imperfect pitchfork

Every coupling moment is $\propto$ the COM offset and vanishes for $a_X = a_Y = 0$; simultaneously M_{aug} becomes diagonal and the leakage channel closes. With a centred COM the strafe-saturation is symmetric and the relevant bifurcation is a pitchfork; the COM offset is the imperfection parameter that unfolds it into a saddle-node, preselecting the yaw direction.

6 Post-cap behaviour: yaw asymmetry and cross-axis leak

6.1 Directional (CW/CCW) asymmetry of the yaw cap

Every friction drag is even in $\dot{\psi}$; the COM offset supplies the only odd-in- $\dot{\psi}$ coupling, via (i) a fixed centripetal $-m(a_X, a_Y)\dot{\psi}^2$ added to a sign-flipping tangential demand $\pm\mu N_i \hat{t}_i$, which add at one spin sense and cancel at the other, and (ii) the gyroscopic yaw term $m\dot{\psi}|a|(\hat{a} \cdot \vec{V})$. The asymmetry scales as $m|a|\dot{\psi}^2/(4\mu N_{\min})$ — $\approx 4\%$ at the corrected cap $\dot{\psi}_{\max} = 2.47$ rad/s ($\approx 10\%$ at the old, unreachable friction value 3.8) — and vanishes at $a_X = a_Y = 0$. The motor gate (12) is itself nearly symmetric (the τ_f sign term is the only odd part, ~ 0.1 N m), so it does not materially change this picture.

6.2 Why exceeding $V_{y,crit}$ bleeds into $\dot{\psi}$ and V_x

$V_{x,cap}$ is soft (motor/drag): past it the circle stays empty and the platform merely lags, on-axis and quiet — under the motor model the lag is now a hard torque deficit that grows linearly with the overspeed, but still on-axis. $V_{y,crit}$ is a hard friction-circle wall: past it the passive roller can no longer sustain $F_{\perp}$, $V_{py,i}$ grows, and the saturated force caps and rotates. The load spread breaks symmetry, the yaw moment climbs from 0 at the cap toward 1.5–2.7 N m, forcing $\dot{\psi} \neq 0$; the

induced $\dot{\psi}$ re-enters V_x through the Coriolis $-m_s\dot{\psi}V_y$ and the $-ma_Y$ off-diagonal. With d_y lost,

$$\tilde{m}\dot{V}_x = -d_x V_x + m_s\dot{\psi}V_y, \quad \tilde{m}\dot{V}_y = -\underbrace{d_y V_y}_{\rightarrow 0} - m_s\dot{\psi}V_x, \quad (54)$$

so the skew $m_s\dot{\psi}$ whirls V_y into V_x and back, a bounded-then-diverging lateral/yaw oscillation. A longitudinal over-command does none of this — the asymmetry is the soft-vs-hard cap distinction.

6.3 Exceeding $\dot{\psi}_{\max}$: leak into V_x and V_y

The dual of §6.2, with a new leading act. Past the *motor* cap $\dot{\psi}_{\max} = 2.47$ rad/s the wheel voltages sit at the bus limit and back-EMF leaves no torque to arrest error growth: the spin simply plateaus at what the torque-speed line permits, on-axis, and the controller winds up against an authority wall no gain law can cross (the `spiral_orbit` failure mode). If the spin were somehow forced past the friction pinch $\dot{\psi}_{\text{fric}} = 3.8$ rad/s (it cannot be, self-propelled: that lies beyond the no-load wall 3.60 rad/s), the lightest wheel would fail to supply $F_{\perp}$, slide along $\hat{n}$, and deliver only $\mu N_{\min}$; the symmetric balance breaks and a net body force appears,

$$\vec{F}_{\text{leak}} \approx \Delta F \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right), \quad \Delta F = F_{\perp}^{\text{dem}} - \mu N_{\min}, \quad (55)$$

(forward-right for $\dot{\psi} > 0$; flips for $\dot{\psi} < 0$). Since $F_{\parallel} \rightarrow 0$ at the saturated wheels, the traction that drove the spin vanishes ($\dot{\psi}$ plateaus) and the drift is uncorrectable at the very wheels producing it — the likely origin of the observer’s actuator saturation at heading-ramp onset, motivating a per-axis split $\omega_{\psi} \gg \omega_{xy}$.

7 Summary of closed-form results

A Equation appendix — Word-compatible L^AT_EX commands

Each block is the linear L^AT_EX string for the corresponding numbered equation, restricted to commands accepted by the Word equation editor. Concatenate split lines before pasting. Entries (E49)–(E55) are the velocity–acceleration envelope of §3.4, now stated as the two-gate test; (E58)–(E60) are the new actuator equations of §1 and §3.

(E13) Lateral velocity cap (onset, lightest wheel; motor slack verified, Sec 3.2):

$$V_{\text{y,crit}} = \frac{N_{\min}}{\mu}, \quad (R - R_a)^2 \text{ } \{ 2 \text{ p}_2 \}$$

(E33) Yaw-rate caps (three levels, motor binds first, Sec 3.3):

$$\begin{aligned} F_{\perp} &= \frac{\sqrt{2}}{p_2 h} (R - R_a)^2 \backslash, \quad \dot{\psi} \\ \dot{\psi}_{\text{fric}} &= \frac{\mu N_{\min}}{(R - R_a)^2} \sqrt{2} \backslash, \quad p_2 h = 3.8 \text{ (unreachable)} \\ \dot{\psi}_{\text{nl}} &= \omega_{\text{nl}} R / (1 + h) = 3.60 \text{ (no-load kinematic wall)} \\ \dot{\psi}_{\text{max}} &= \tau_{\text{stall}} / (1.10 + 2.60) = 2.47 \text{ rad/s (motor gate, binding)} \\ |F_{\parallel}| &\leq \sqrt{(\mu N_i)^2 - F_{\perp}^2} \text{ (traction headroom)} \end{aligned}$$

(E45) Roller-frame force decomposition (axle e_i , rolling dir n_i ; friction circle):

$$\begin{aligned} \vec{F}_i &= F_{\parallel} \hat{e}_i + F_{\perp} \hat{n}_i, \quad \hat{e}_i \cdot \hat{n}_i = 0 \\ \hat{e}_i &= (\cos \delta_i, \sin \delta_i), \quad \hat{n}_i = (-\sin \delta_i, \cos \delta_i) \\ F_{\parallel}^2 + F_{\perp}^2 &\leq (\mu N_i)^2 \\ F_{\perp} &= p_2 \backslash, \quad \dot{\gamma}_i / (R - R_a) \end{aligned}$$

(E46) Roller-frame yaw arms (traction lever vs roller-drag lever):

Quantity	Expression	Value (case 1, $\mu = 0.5$)
<i>Actuator landmarks (Table 2, §1)</i>		
Stall torque ($V = 24$ V, $\omega = 0$)	$G\eta(K_t V_{\max}/R_a^{(m)} - \tau_f)$	9.12 N m
Torque-speed slope	$G^2\eta K_t K_b/R_a^{(m)}$	0.337 N m s/rad
No-load wheel speed (hard wall)	$V_{\max}/(K_b G)$	27.73 rad/s
<i>Velocity caps (§3.1–3.3, motor-aware)</i>		
Longitudinal speed cap	$\tau_{avail}(\omega) = p_1 \omega \text{ root}, \times R$	1.02 m/s (73.5% nl)
hard ceiling (no-load)	$\omega_{nl} R$	1.386 m/s
Lateral onset (1st wheel, friction)	$N_{\min} \mu (R - R_a)^2 / (2p_2)$	0.63 m/s (motor slack)
Lateral full saturation	$(\sum N_i) \mu (R - R_a)^2 / (8p_2)$	0.79 m/s
Yaw-rate cap (motor-limited)	$\tau_{avail}(7.7\dot{\psi}) = 1.10 \dot{\psi} \text{ root}$	2.47 rad/s (68.5% nl)
yaw no-load ceiling	$\omega_{nl} R / (l + h)$	3.60 rad/s
yaw friction pinch (unreachable)	$N_{\min} \mu (R - R_a)^2 / (\sqrt{2} p_2 h)$	3.8 rad/s
Longitudinal accel. cap (aggregate)	$(\sum N_i) \mu / m_s$	4.9 m/s ²
<i>Velocity-acceleration envelope (§3.4, per-wheel, bare mass, RL-bound)</i>		
Long. accel. cap (per-wheel, at rest)	$(\mu N_3) \text{ free at } \vec{v} = 0$	2.93 m/s ²
Lateral accel. cap (per-wheel, at rest)	$(\mu N_3) \text{ free at } \vec{v} = 0$	2.86 m/s ²
Yaw accel. cap (per-wheel, at rest)	$(\mu N_3) \text{ free at } \vec{v} = 0$	9.62 rad/s ²
Stall traction headroom (motor gate)	τ_{stall}/R	182 N/wheel (slack at rest)
Long. traction headroom vs speed	$(\tau_{avail} - p_1 \omega)/R$	182.3 – 66.0 V_x N
Drag-relocation $\Phi(\vec{v})$	$(0, 4p_2 V_y / (R - R_a)^2, 2.2\dot{\psi})$	$(0, 110 V_y, 2.2\dot{\psi})$
Master envelope (two gates)	friction (28) + motor (12)	—
<i>Out-of-plane load-transfer bound (§3.5)</i>		
Per-wheel load transfer	$m a z_{com} / (4d)$	$d = h \text{ long.}, l \text{ lat.}$
Operating accel. cap (<5% transfer)	$0.05 \cdot 4l N_3 / (m z_{com})$	1.0 m/s ² ($z_{com} = R$)
Contact yaw moment of $F_{\perp}$	$4p_2 h (h - l) / (R - R_a)^2$	≈ 2.2 N m s
CW/CCW cap asymmetry (at $\dot{\psi}_{\max} = 2.47$)	$m a \dot{\psi}^2 / (4\mu N_{\min})$	$\approx 4\%$
Yaw centrifugal threshold	$\sqrt{\mu N_{tot} / (m a)}$	14 rad/s
Anisotropy ratio	$V_{x,cap} / V_{y,crit}$	≈ 1.6
Sustained yaw moment (lat.)	ramp $0 \rightarrow \mu m g a_X (h + l) / (\sqrt{2} h)$	0 – 1.5–2.7 N m
Transient yaw moment (lon.)	$\mu m g a_Y (h + l) / (\sqrt{2} l)$	6.94 N m

$\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}, \lambda_i, \text{quad } \lambda = (-, +, -, +)$
 $\ell_{\perp,i} = \frac{h-l}{\sqrt{2}}, \mu_i, \text{quad } \mu = (+, +, -, -)$

===== velocity-acceleration envelope (Sec 3.4, two gates) =====

(E49) Per-wheel acceleration headroom (master velocity-acceleration envelope, TWO gates):

(friction) $(A_{\parallel}^i)^2 + F_{\perp,i}(\vec{v})^2 \leq (\mu N_i)^2$
(motor) $R [(A_{\parallel}^i \cos \delta_i - F_{\perp,i} \sin \delta_i) + p_1 \omega_i + J_w \dot{\omega}_i \leq \tau_{avail}(\omega_i), \text{quad } i = 1..4$
 $\vec{b}(\vec{v}, \dot{\vec{v}}) = M, \dot{\vec{v}} + C(\dot{\psi}), \vec{v} - A_n, \vec{F}_{\perp}$

(E50) Free-component (traction) allocator and its min-norm pseudo-inverse:

$A_{\parallel} = \begin{bmatrix} \cos \delta_1 & \sin \delta_1 & \rho_X \sin \delta_1 & -\rho_Y \sin \delta_1 \\ \cos \delta_2 & \sin \delta_2 & \rho_X \sin \delta_2 & -\rho_Y \sin \delta_2 \end{bmatrix}$
 $\ker A_{\parallel} = \text{tfrac{1}{2}}(1, 1, -1, -1), \text{quad } \vec{F}_{\parallel} = A_{\parallel}^+ \vec{b}$
(squeeze null vector loads all four friction circles AND all four motors: doubly avoided)

(E51) Drag-relocation operator and closed-form offset (velocity-only, linear):

$$A_n = \begin{bmatrix} -\sin\delta_i & \cos\delta_i \\ \cos\delta_i & \sin\delta_i \end{bmatrix} \begin{bmatrix} \rho_{Xi} \cos\delta_i + \rho_{Yi} \sin\delta_i \\ \rho_{Xi} \sin\delta_i - \rho_{Yi} \cos\delta_i \end{bmatrix}$$

$$\Phi(\vec{v}) = A_n \vec{F}_{\perp} = \left(0, \frac{4 p_2}{(R-R_a)^2} V_y, \frac{4 p_2 h}{(R-R_a)^2} \dot{\psi} \right)$$

$$= (0, 110.0 V_y, 2.20 \dot{\psi})$$

(E52) Bare-vs-augmented mass caution (circle uses bare M; motor sees M_{aug} AND J_w):

$$A_{\parallel} \vec{F}_{\parallel} = M \ddot{x} + \dots \quad M = \text{diag}\{\text{block}(m_s, m_s, I_s)\}$$

\text{with COM coupling}

\text{wheel inertia } 4 J_w / R^2 \text{ to } \text{motor gate via } J_w \dot{\omega}_i \text{ (E49 motor row), NOT contact circle}

(E53) Acceleration intercepts from rest ($v = 0$, $F_{\perp} = 0$, full circle free, stall torque):

$$a_{x, \text{cap}} = 2.93, \quad a_{y, \text{cap}} = 2.86 \text{ m/s}^2, \quad \ddot{\psi}_{\text{cap}} = 9.62 \text{ rad/s}^2$$

\text{all pinched at } N_{\min} \text{ (rear-left, friction gate)}, \quad \mu g = 4.9 \text{ m/s}^2

\text{motor gate slack at rest: } \tau_{\text{stall}} / R = 182 \text{ N/wheel} \quad \mu N_3 = 34.8 \text{ N}

(E54) Binding-wheel scalar gate PAIR (friction: wheel #3; motor: $\arg\max |\omega_i|$ wheel j):

$$\kappa = \frac{1}{\sqrt{2}} \sqrt{(R - R_a)^2} = 38.9$$

$$F_{\perp, 3} = \kappa (V_y - h \dot{\psi})$$

$$F_{\parallel, 3} = 0.354 (b_x + b_y) - 0.918, \quad b_{\Omega}$$

(friction) $F_{\parallel, 3}^2 \leq (\mu N_3)^2 - F_{\perp, 3}^2, \quad \mu N_3 = 34.8$

(motor) $R (F_{\parallel, j} \mp F_{\perp, j}) / \sqrt{2} + p_1 \omega_j + J_w \dot{\omega}_j \leq \tau_{\text{avail}}(\omega_j), \quad |\omega_i| < \omega_{\text{nl}} = 27.73 \text{ rad/s} \quad \forall i$

(E55) Demand vector components $b(v, \dot{v})$ for the gate ($\dot{v} = (a_x, a_y, \alpha)$):

$$b_x = m_s a_x - m a_Y \alpha - m_s \dot{\psi} V_y - m a_X \dot{\psi}^2$$

$$b_y = m_s a_y + m a_X \alpha + m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 - 110.0, \quad V_y$$

$$b_{\Omega} = -m a_Y a_x + m a_X a_y + I_s \alpha + m \dot{\psi} (a_X V_x + a_Y V_y) - 2.20 \dot{\psi}$$

(E56) Factor of safety (applied to the whole circle, both components):

$$\mu N_i \geq s, \quad \mu N_i, \quad s \in [0.7, 0.8]$$

(E57) Out-of-plane per-wheel load transfer (sets the $a \leq 1.0 \text{ m/s}^2$ operating cap):

$$\Delta N_i = \frac{m a_i}{z_{\text{com}}} \quad d = h \text{ (long.)} \quad l \text{ (lat.)}$$

$$a_i^{\text{lat}} = \frac{0.05 \cdot 4 l N_3}{m z_{\text{com}}} = 1.39 \text{ (z=R)} \quad 0.70 \text{ (z=2R)}$$

===== actuator / back-EMF (new, Sec 1 and Sec 3) =====

(E58) Quasi-static DC-motor map and torque-speed line (plant.jl motor_torque):

$$i = (V - K_b G \omega) / R_a(m), \quad M = G \eta (K_t i - \tau_f \text{sign}(\omega))$$

$$\tau_{\text{avail}}(\omega) = G \eta (K_t V_{\text{max}} / R_a(m) - \tau_f) - (G^2 \eta K_t K_b / R_a(m)) |\omega|$$

$$= 9.12 - 0.337 |\omega| \text{ N}\cdot\text{m}$$

$$\omega_{\text{nl}} = V_{\text{max}} / (K_b G) = 27.73 \text{ rad/s (no-load, hard wall)}$$

(voltage clamp 24 V / 2 Ohm = 12 A binds before $i_{\text{max}} = 12.8 \text{ A}$: current limit slack)

(E59) Longitudinal cap from the torque-speed line (Sec 3.1; supersedes $\tau_{\text{max}} R / p_1 = 4.55$):

$$\omega_* = \frac{G \eta (K_t V_{\text{max}} / R_a(m) - \tau_f)}{p_1 + G^2 \eta K_t K_b / R_a(m)}$$

$$= 20.39 \text{ rad/s}$$

$$V_{x, \text{cap}} = \omega_* R = 1.02 \text{ m/s} \quad (= 73.5\% \text{ no load})$$

$$F_{\text{head}}(V_x) = (\tau_{\text{avail}}(V_x/R) - p_1 V_x/R) / R = 182.3 - 66.0 V_x \text{ N/wheel}$$

(E60) No-slip wheel speed / acceleration maps (motor-gate inputs, Sec 3.4):

$$\begin{aligned}\omega_i &= \frac{V_x + \tan\delta_i V_y + (\rho_{Xi} \tan\delta_i - \rho_{Yi}) \dot{\psi}}{R} \\ &= (V_x \pm V_y \pm (1+h) \dot{\psi}) / R \\ \dot{\omega}_i &= (a_x \pm a_y \pm (1+h) \alpha) / R\end{aligned}$$

Reference. Adamov, B.I., & Saipulaev, G.R. (2020). Research on the dynamics of an omnidirectional platform taking into account real design of mecanum wheels (as exemplified by KUKA youBot). *Russian Journal of Nonlinear Dynamics*, 16(2), 291–307.